



GRAPH-BASED DETECTION, SEGMENTATION & CHARACTERIZATION OF BRAIN TUMORS

**A Spatial Prior Approach for Low-
Grade Gliomas**

AUTHERS

Sarah Parisot

Hugues Duffau

Stephane Chemouny

Nikos Paragios

GROUP MEMBERS

**Akshayan K.
EG/2021/4393**

**Dissanayake W.M.N.P.
EG/2021/4499**

**Shageethpratheep V.
EG/2021/4809**

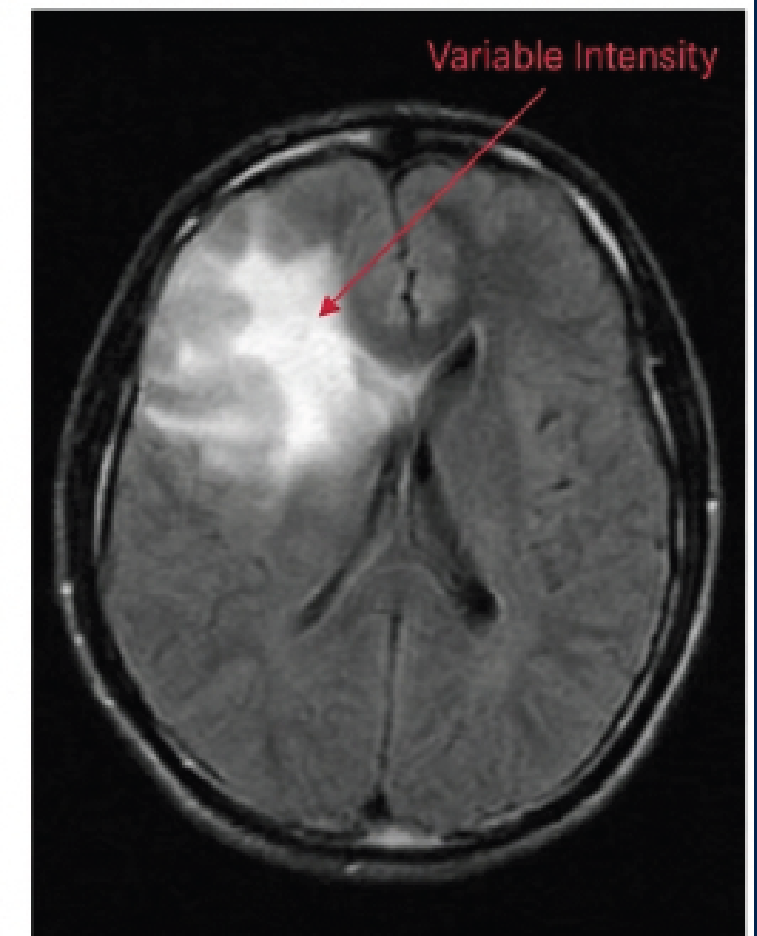
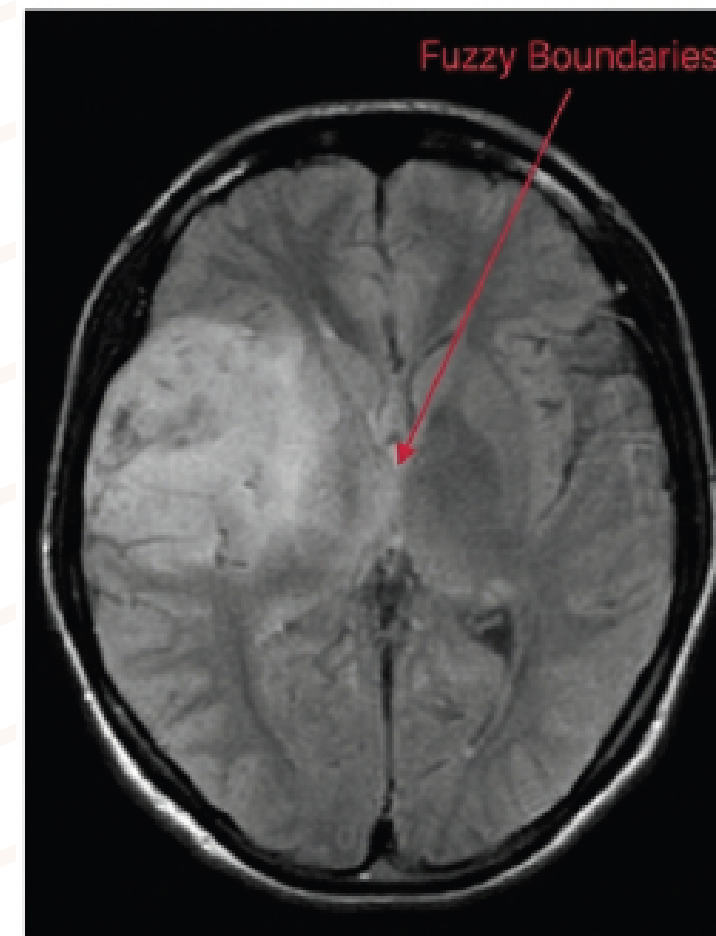
**Shayanthavi T.
EG/2021/4811**

MOTIVATION & PROBLEM STATEMENT

- Clinical Goal: Early detection and segmentation of Low-Grade Gliomas (LGG).

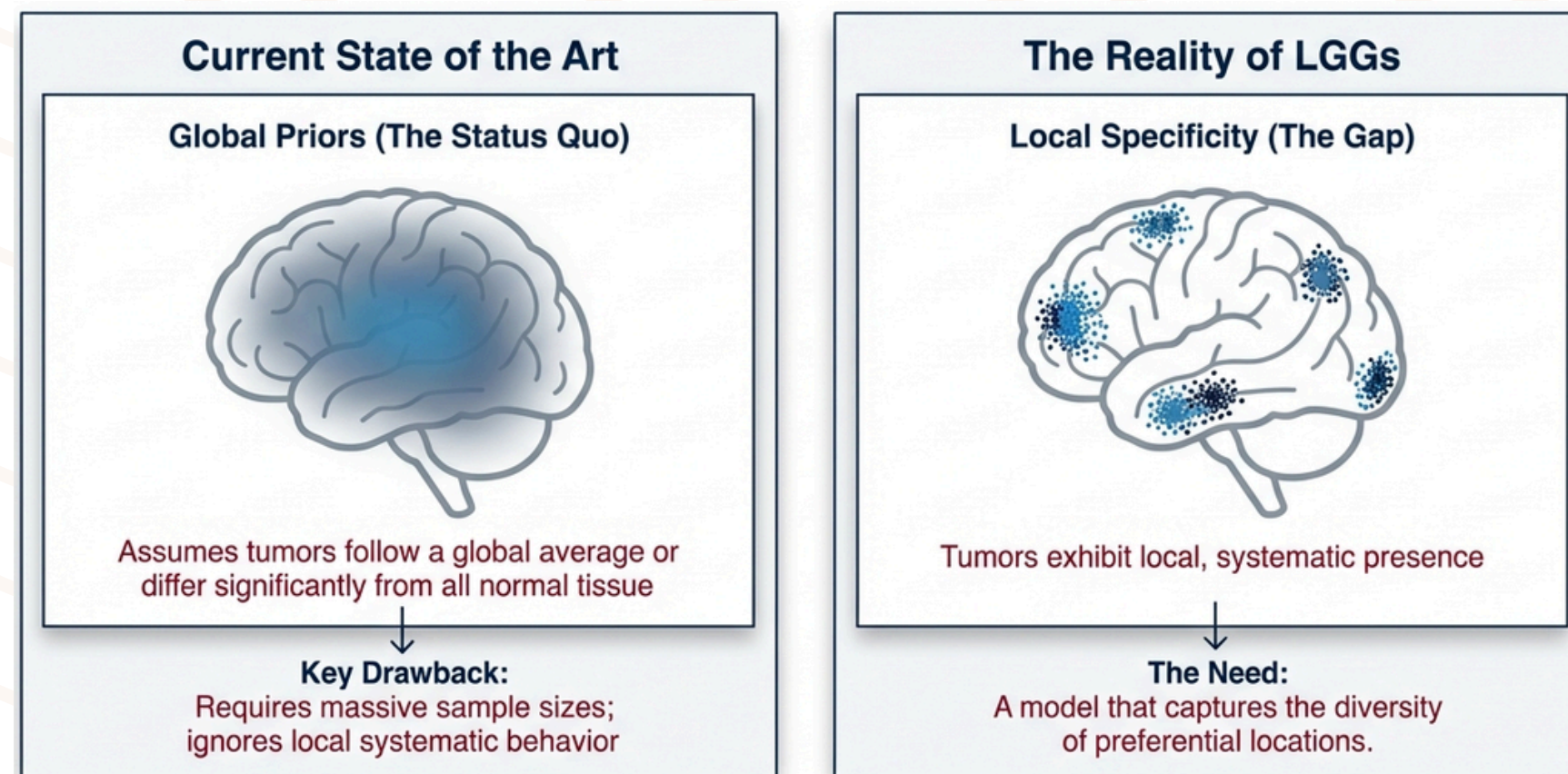
The Challenge:

- LGGs have high variability in shape, size, and location.
 - They often exhibit fuzzy boundaries and low contrast against healthy tissue.
- The Gap in Existing Research:
 - Existing methods rely on Global Priors (standard atlases).
 - Limitation: Global priors fail to capture the specific, localized deformation patterns of tumors.



THE AUTHORS' CORE HYPOTHESIS

- Preferential Locations: The paper builds on the medical finding that LGGs are not random; they occur in predictable "preferential locations".
- Proposed Solution:
 - Move from a Global Prior to a Localized Sparse Graph.
 - By clustering tumors based on spatial proximity, the system can predict where a specific subtype of tumor is likely to appear.



METHODOLOGY

1. GRAPH CONSTRUCTION

- Dataset: 113 manually segmented MRI scans.
- Similarity Metric: The authors calculate the Mahalanobis Distance between tumors to measure spatial overlap:

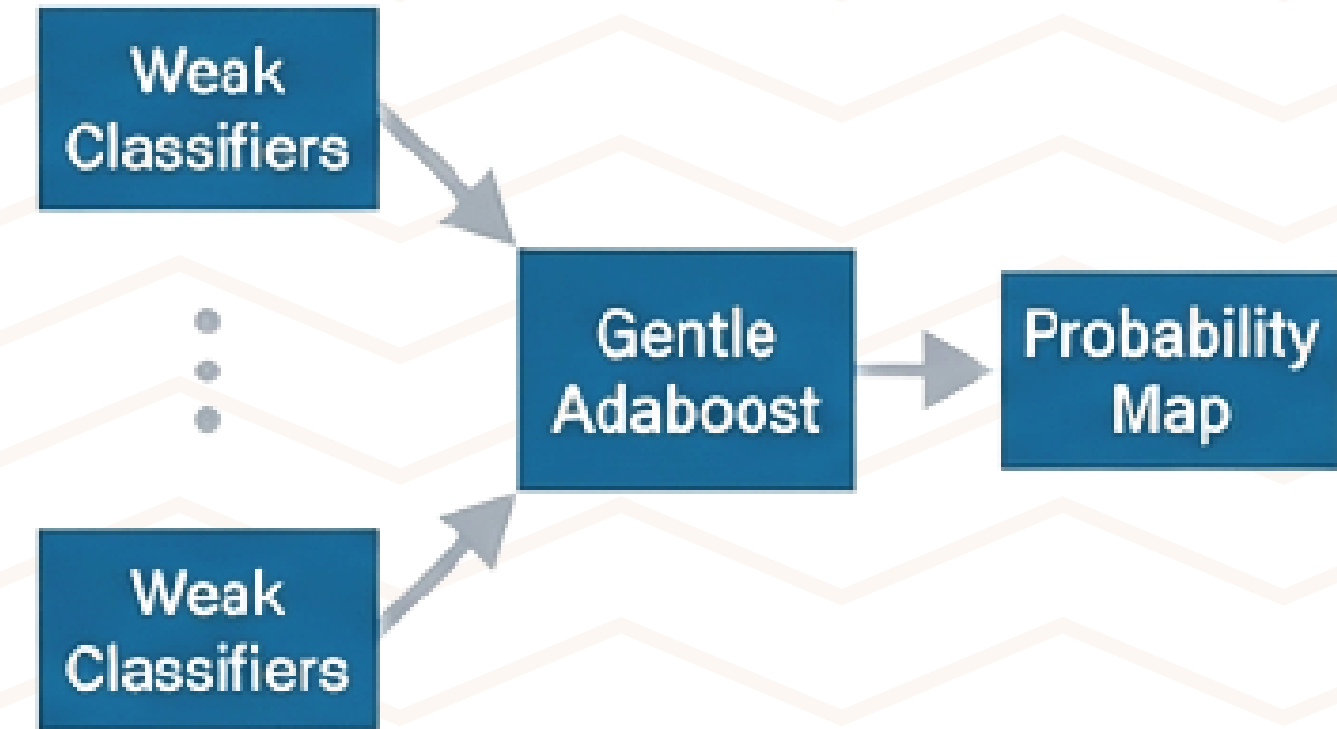
$$d_M(S_i, S_j) = \sqrt{(\bar{x}_i - \bar{x}_j)^T \Sigma^{-1} (\bar{x}_i - \bar{x}_j)}$$

- Clustering:
 - Unsupervised clustering groups tumors into distinct nodes (classes).
 - Result: A graph where each node represents a "type" of tumor location.

METHODOLOGY

2. VISUAL DETECTION (BOOSTING)

- Classifier: Gentle AdaBoost algorithm.
- Feature Engineering:
 - Intensity: Patch-based statistics to handle inhomogeneity.
 - Texture: Gabor features (multiscale).
 - Symmetry: A custom feature measuring dissymmetry across the brain's midline
- Output: A probability map indicating likelihood of tumor presence per voxel.

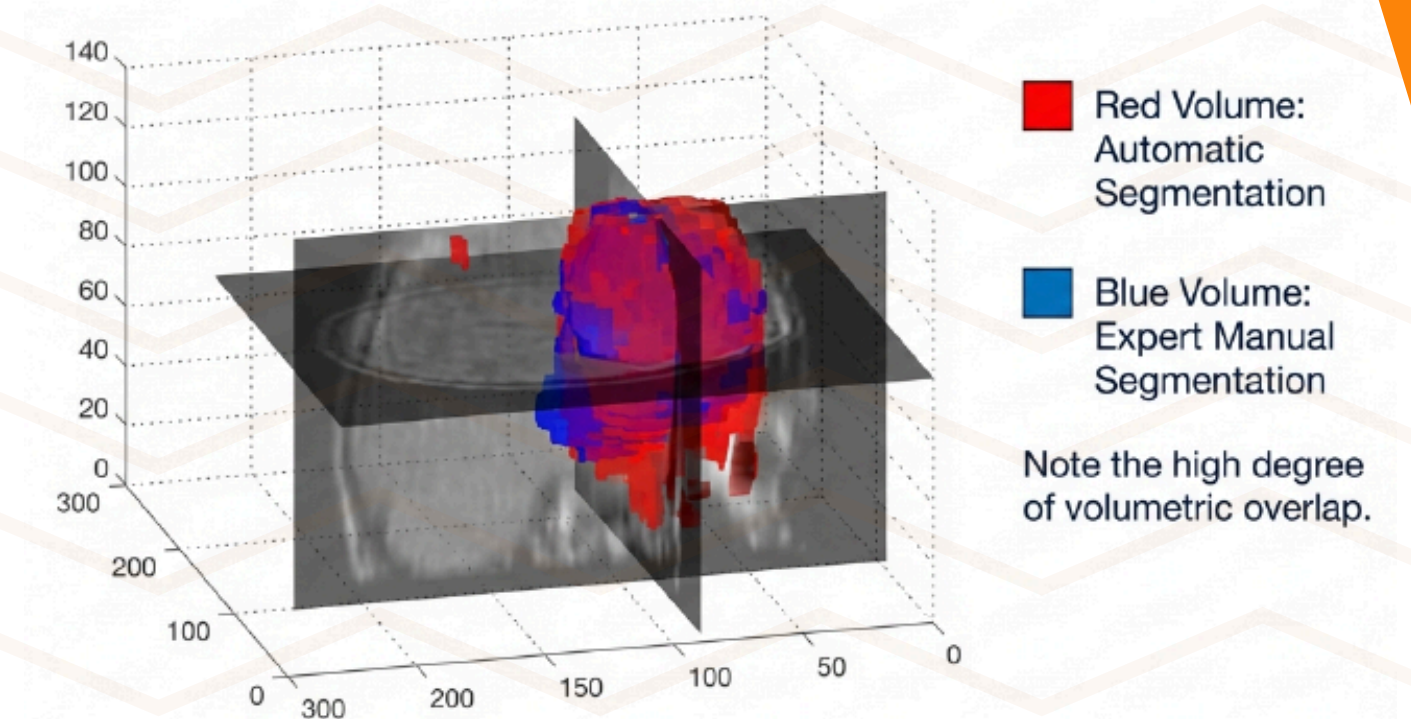


EXPERIMENTAL SETUP

- Data: 113 MRI FLAIR images (Low-Grade Gliomas).
- Split: 40 images for training / 73 images for testing.
- Preprocessing:
 - Affine registration to a reference atlas.
 - Intensity normalization using Median and IQR.

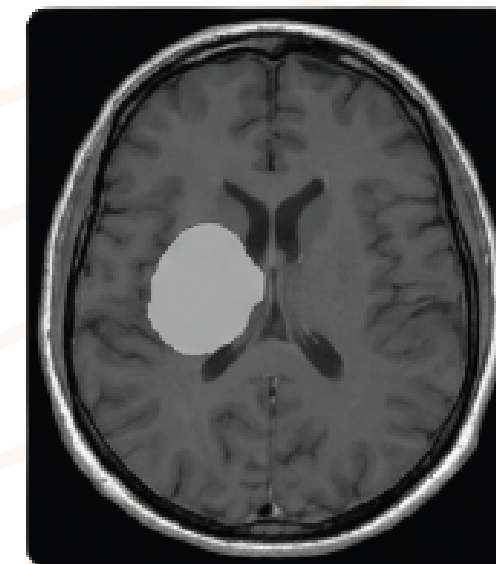
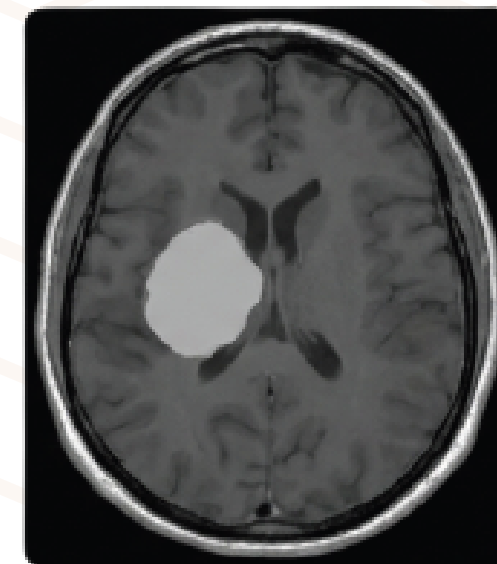
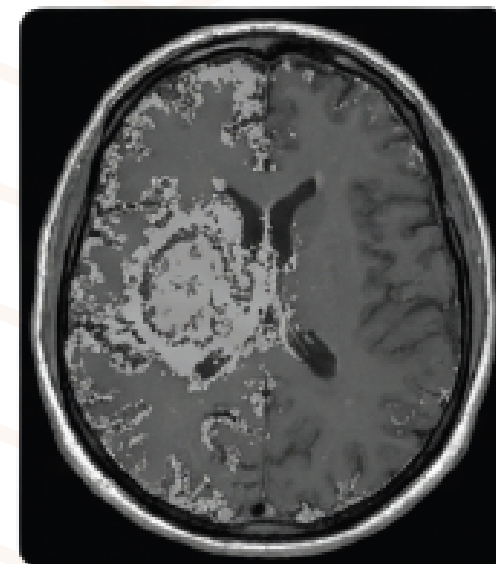
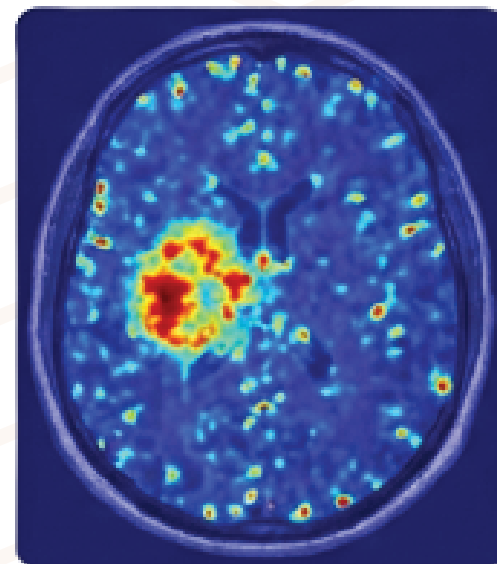
QUANTITATIVE RESULTS

- Key Metric: Dice Score (Segmentation Accuracy).
 - Baseline (Boosting only): 48%
 - Standard MRF (Regularization): 65%
 - Proposed Method (With Prior): 74%
- False Positive Rate:
 - Reduced from 45% (baseline) to 24% (proposed method).



QUANTITATIVE RESULTS (VISUALS)

- Successes:
 - The method successfully cleans up "noisy" classifications from the boosting step.
 - Robust to low contrast and fuzzy boundaries common in LGGs.
- False Positive Rate:
 - Large Tumors: The prior fails for very large tumors (>200 cubic cm).
 - Reason: Large tumors span the spatial domains of multiple clusters, confusing the model.



CRITICAL ANALYSIS

- Strengths:
 - Novelty: Innovative use of "preferential locations" rather than a generic average atlas.
 - Integration: Effectively combines machine learning (AdaBoost) with geometric priors (Graphs).
 - Performance: Significant reduction in false positives (almost 50% reduction)

CRITICAL ANALYSIS

- Weaknesses/Critique:
 - Dependency: Heavily relies on accurate affine registration to the atlas.
 - Scope: Specifically designed for Low-Grade Gliomas; may not generalize to other tumor types with random locations (e.g., glioblastomas).
 - Complexity: Requires training a complex graph structure; computational load is higher than simple thresholding.

CONCLUSION

- Summary: Parisot et al. present a robust framework for joint segmentation and characterization.
- Key Takeaway: Incorporating localized spatial knowledge (where tumors usually are) significantly aids purely visual detection (what tumors look like).
- Future Impact: Potential for monitoring tumor progression or application to other organ pathologies.

**THANK
YOU**